Problem Set 1 by Jimmy Zheng

This is an R Markdown Notebook. When you execute code within the notebook, the results appear beneath the code.

Try executing this chunk by clicking the *Run* button within the chunk or by placing your cursor inside it and pressing *Cmd+Shift+Enter*.

```
x <- c(1, 3, 2, 5)
x
```

```
## [1] 1 3 2 5
```

```
x= c(1, 6, 2)
y= c(1, 4, 3)
```

```
length (x)
```

```
## [1] 3
```

```
length (y)
```

```
## [1] 3
```

```
x+y
```

```
## [1]  2 10  5
```

```
ls()
```

```
## [1] "x" "y"
```

```
rm(x, y)
ls()
```

```
## character(0)
```

```
rm(list = ls ())
```

```
?matrix
```

```r
x <- matrix (data = c(1, 2, 3, 4), nrow = 2, ncol = 2)
x
```

```
##      [,1] [,2]
## [1,]    1    3
## [2,]    2    4
```

```r
 x <- matrix (c(1, 2, 3, 4), 2, 2)
x
```

```
##      [,1] [,2]
## [1,]    1    3
## [2,]    2    4
```

```r
matrix (c(1, 2, 3, 4), 2, 2, byrow = TRUE)
```

```
##      [,1] [,2]
## [1,]    1    2
## [2,]    3    4
```

```r
sqrt (x)
```

```
##          [,1]     [,2]
## [1,] 1.000000 1.732051
## [2,] 1.414214 2.000000
```

```r
x^2
```

```
##      [,1] [,2]
## [1,]    1    9
## [2,]    4   16
```

```r
x <- rnorm (50)
y <- x + rnorm (50, mean = 50, sd = .1)
cor (x, y)
```

```
## [1] 0.9954009
```

```r
set.seed (1303)
rnorm (50)
```

```
##  [1] -1.1439763145  1.3421293656  2.1853904757  0.5363925179  0.0631929665
##  [6]  0.5022344825 -0.0004167247  0.5658198405 -0.5725226890 -1.1102250073
## [11] -0.0486871234 -0.6956562176  0.8289174803  0.2066528551 -0.2356745091
## [16] -0.5563104914 -0.3647543571  0.8623550343 -0.6307715354  0.3136021252
## [21] -0.9314953177  0.8238676185  0.5233707021  0.7069214120  0.4202043256
## [26] -0.2690521547 -1.5103172999 -0.6902124766 -0.1434719524 -1.0135274099
## [31]  1.5732737361  0.0127465055  0.8726470499  0.4220661905 -0.0188157917
## [36]  2.6157489689 -0.6931401748 -0.2663217810 -0.7206364412  1.3677342065
## [41]  0.2640073322  0.6321868074 -1.3306509858  0.0268888182  1.0406363208
## [46]  1.3120237985 -0.0300020767 -0.2500257125  0.0234144857  1.6598706557
```

```r
set.seed (3)
y <- rnorm (100)
```

```r
mean (y)
```

```
## [1] 0.01103557
```

```r
var (y)
```

```
## [1] 0.7328675
```

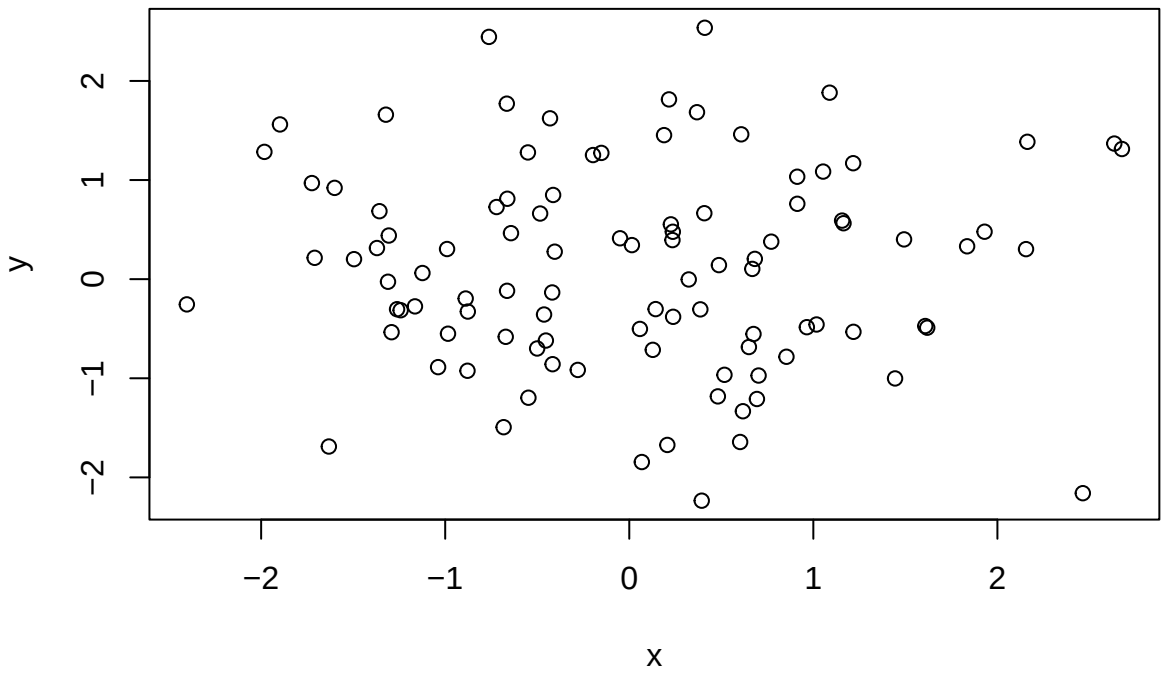
```r
sqrt ( var (y))
```

```
## [1] 0.8560768
```

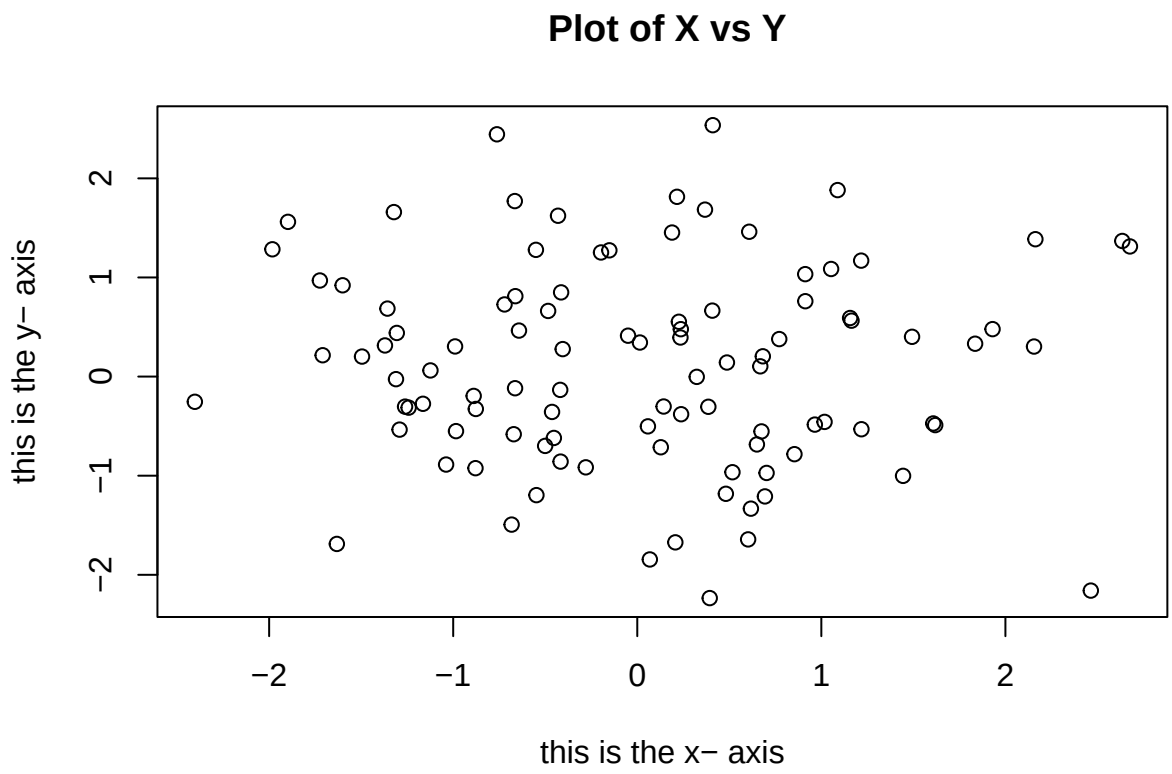
```r
sd(y)
```

```
## [1] 0.8560768
```

```r
x <- rnorm (100)
y <- rnorm (100)
plot (x, y)
```



```
plot (x, y, xlab = " this is the x- axis ",
ylab = " this is the y- axis ",
main = " Plot of X vs Y")
```




```
pdf (" Figure . pdf ")
plot (x, y, col = " green ")
dev.off ()
```

```
## pdf
##   2
```

```
x <- seq (1, 10)
x
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

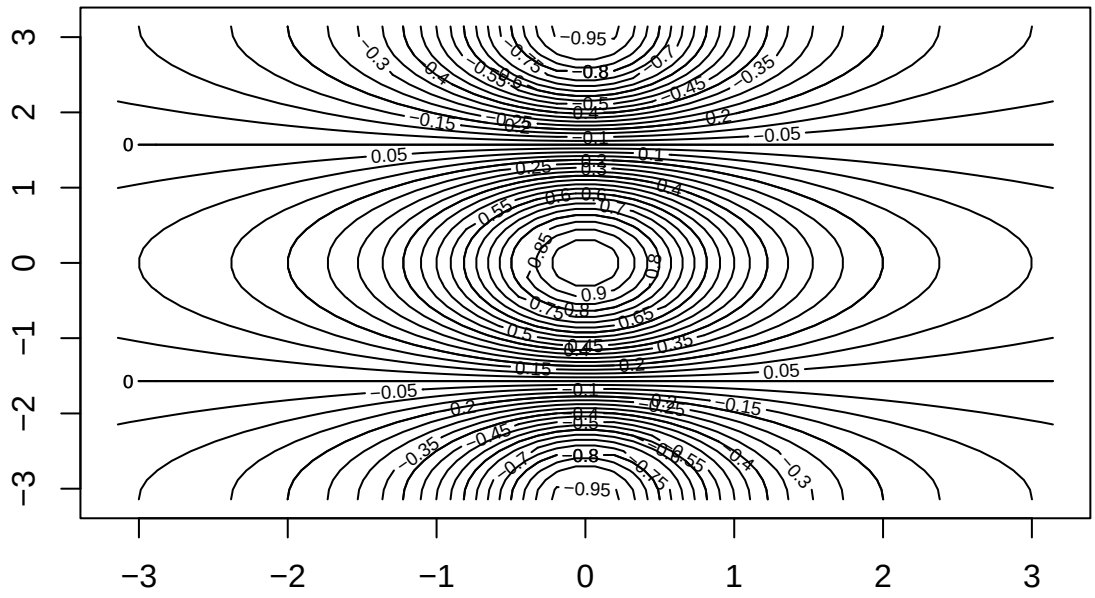
```
x <- 1:10
x
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

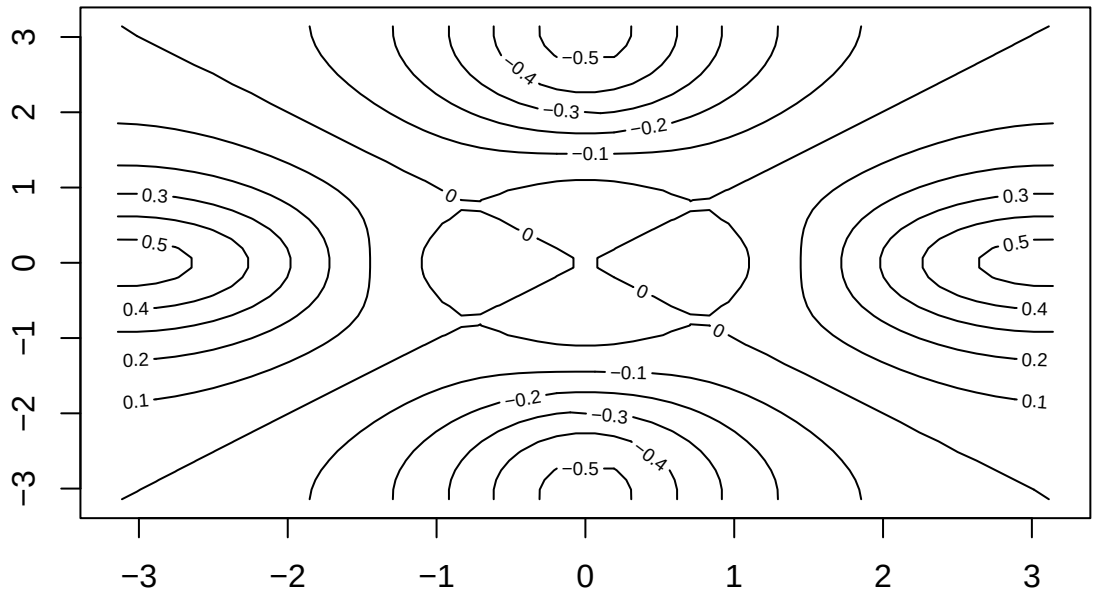
```
x <- seq (-pi, pi, length = 50)
x
```

```
##  [1] -3.14159265 -3.01336438 -2.88513611 -2.75690784 -2.62867957 -2.50045130
##  [7] -2.37222302 -2.24399475 -2.11576648 -1.98753821 -1.85930994 -1.73108167
## [13] -1.60285339 -1.47462512 -1.34639685 -1.21816858 -1.08994031 -0.96171204
## [19] -0.83348377 -0.70525549 -0.57702722 -0.44879895 -0.32057068 -0.19234241
## [25] -0.06411414  0.06411414  0.19234241  0.32057068  0.44879895  0.57702722
## [31]  0.70525549  0.83348377  0.96171204  1.08994031  1.21816858  1.34639685
## [37]  1.47462512  1.60285339  1.73108167  1.85930994  1.98753821  2.11576648
## [43]  2.24399475  2.37222302  2.50045130  2.62867957  2.75690784  2.88513611
## [49]  3.01336438  3.14159265
```

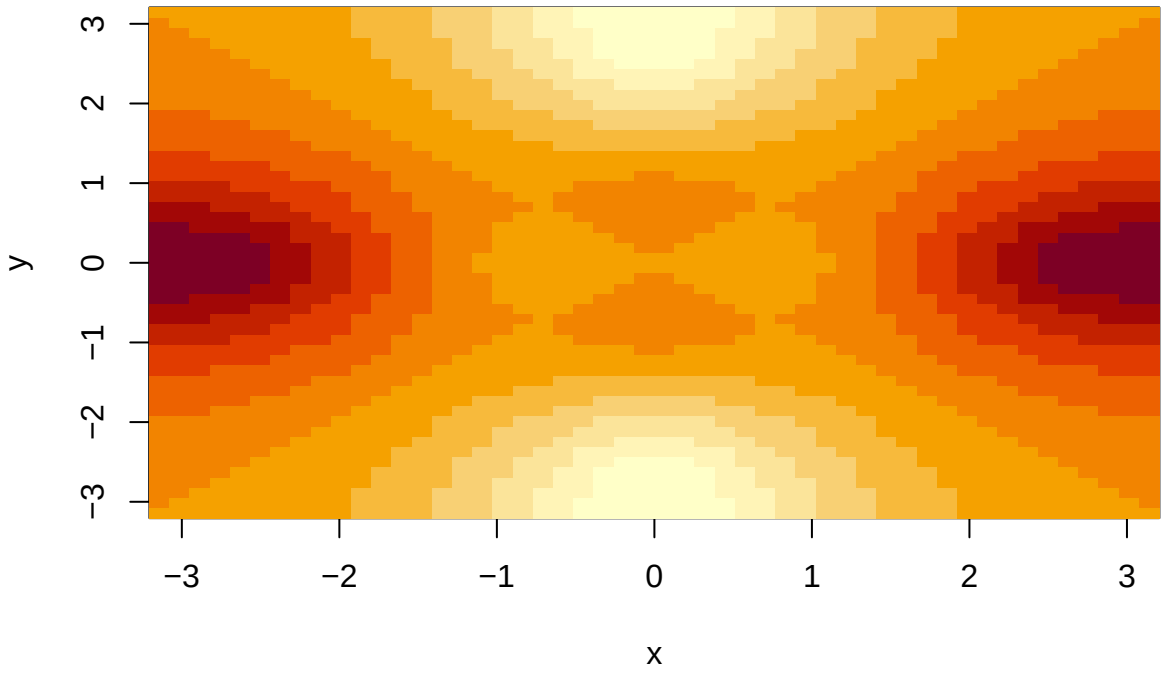
```
y <- x
f <- outer (x, y, function (x, y) cos (y) / (1 + x^2))
contour (x, y, f)
contour (x, y, f, nlevels = 45, add = T)
```



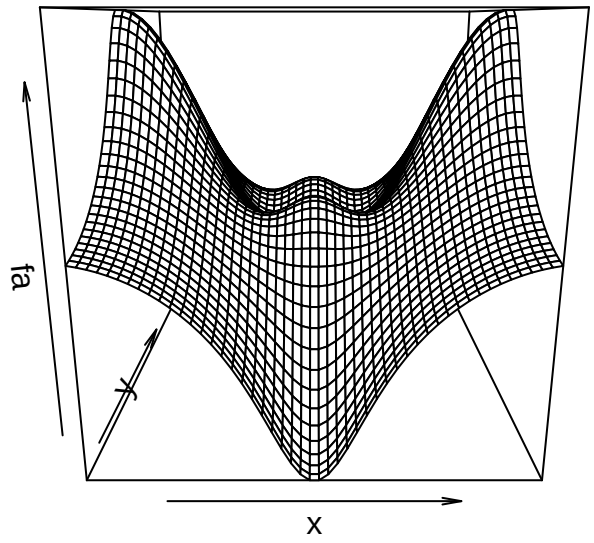

```
fa <- (f - t(f)) / 2
contour (x, y, fa, nlevels = 15)
```



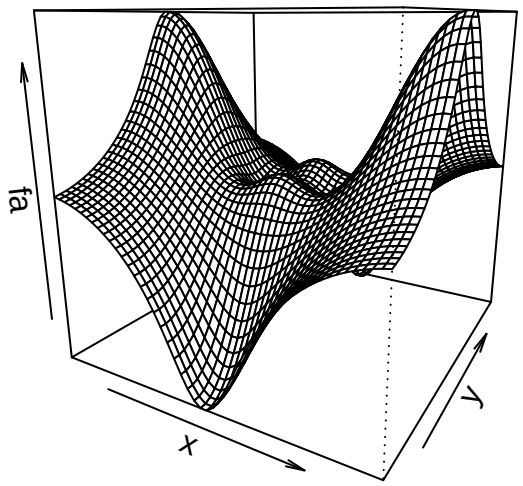

```
image (x, y, fa)
```



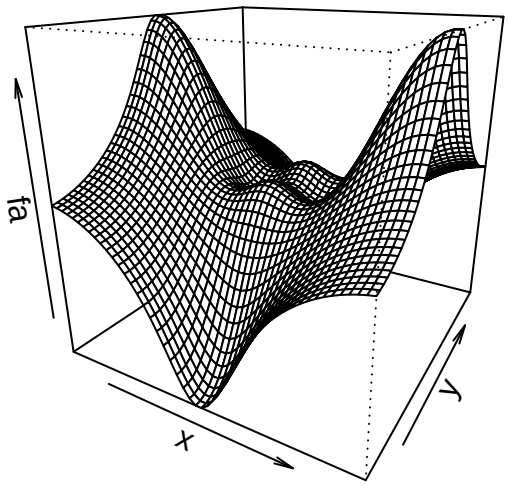

```
persp (x, y, fa)
```



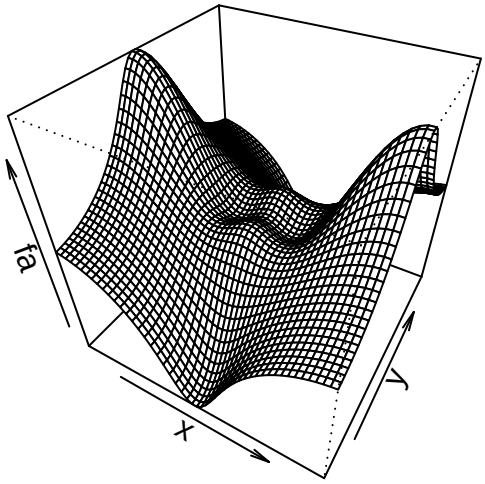

```
persp (x, y, fa , theta = 30)
```



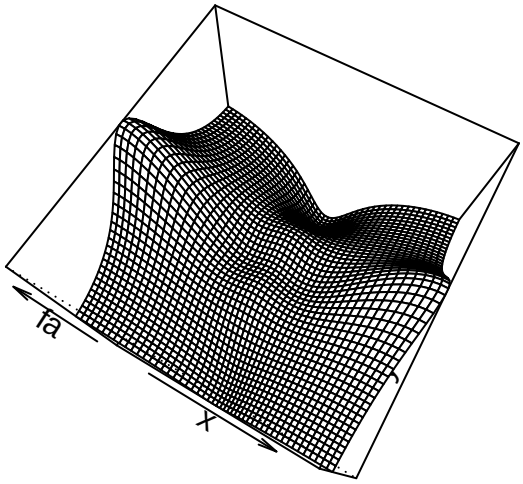

```
persp (x, y, fa , theta = 30, phi = 20)
```




```
persp (x, y, fa , theta = 30, phi = 40)
```




```
persp (x, y, fa , theta = 30, phi = 70)
```


fa
x
r